

Equity Valuation Casebook for MBA Students: Students' Edition

Equity Valuation Casebook for MBA Students

Dr. Srikanth Potharla

Purpose of the Casebook

Intended Audience

Learning Approach

Instructions for Students

Organization of the Casebook

?	?
?	
?	
?	
?	
?	
?	?
?	
?	
?	

Symbols and Abbreviations

₹:

EPS: ?

ROE:

Ke:

WACC:

g:

b:

DDM:

FCFF:

FCFE:

EBIT:

EBITDA:

NOPAT:

PVGO:

P/E:

P/B:

P/S: 

P/CF:

PEG:

EV:

APV:

EVA:

MVA:

LBO:

IRR:

CMP:

cr:

General Rounding Convention

Table of Contents

?

Error! Bookmark not defined.

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

Error! Bookmark not defined.

Error! Bookmark not defined.

Error! Bookmark not defined.

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

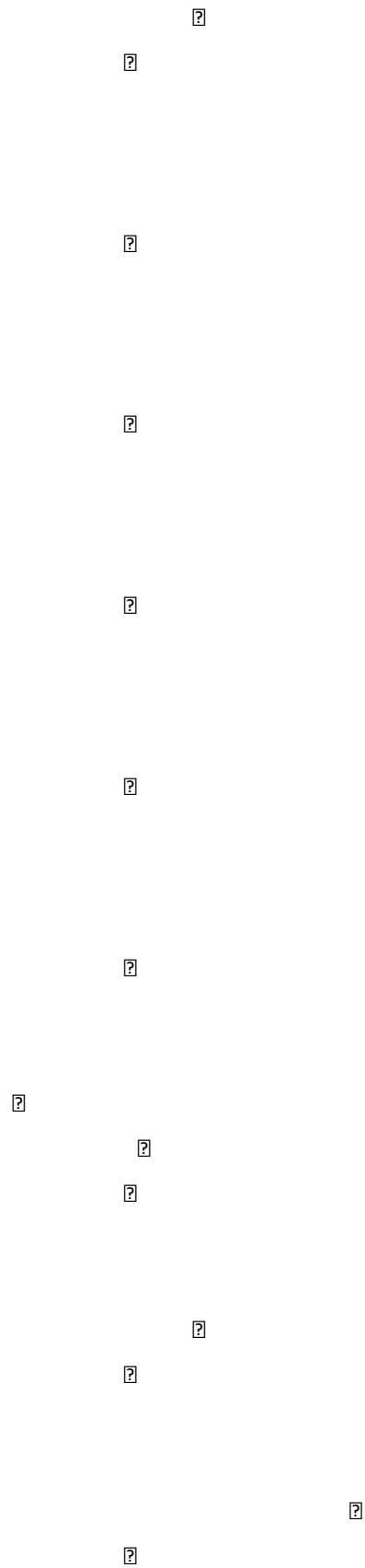
?

?

?

?

?



?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

SECTION I

Valuation Foundations and Financial Statement Preparation

Case 1: Intrinsic Value versus Market Price

Learning Objective:

Case Situation

?

?

Required

Final Answer

Case 2: Equity Value versus Enterprise Value

Learning Objective:

Case Situation

Required

Final Answer

Case 3: Normalising Reported Earnings

Learning Objective:

Case Situation

Required

Final Answer

Case 4: Identifying Sustainable Operating Cash Flow

Learning Objective:

Case Situation

Required

Final Answer

Case 5: Reconstructing Free Cash Flow from Financial Statements

Learning Objective:

Case Situation

?

Required

Final Answer

Case 6: Matching Cash Flow and Discount Rate

Learning Objective:

Case Situation

Required

?

Final Answer

SECTION II

Dividend Discount Models and Growth Opportunities

Case 7: Stable-Growth Dividend Discount Model

Learning Objective:

Case Situation



Required

Final Answer

Case 8: Sustainable Growth in the DDM

Learning Objective:

Case Situation



Required

Final Answer

?

?

Case 9: Present Value of Growth Opportunities

Learning Objective: ?

Case Situation

?

?

Required

Final Answer

Case 10: Retention Ratio and Shareholder Value

Learning Objective:

Case Situation

?

Required

Final Answer

☐ ☐ ☐ ☐

Case 11: Effect of a Change in Dividend Payout

Learning Objective:

Case Situation

Required

Final Answer

Case 12: Two-Stage Dividend Discount Model

Learning Objective:

Case Situation

Required

Final Answer

Case 13: High-Growth Phase Followed by a Transition Period

Learning Objective:

Case Situation

Required

Final Answer

Case 14: Three-Stage Dividend Discount Model

Learning Objective:

Case Situation

Required

Final Answer

Case 15: EPS, Payout, and Multistage Dividends

Learning Objective:



Case Situation



Required



Final Answer



Case 16: DDM Suitability for an Irregular Dividend Company

Learning Objective:

Case Situation

Required

Final Answer

Case 17: Market Price, DDM Value, and Recommendation Risk

Learning Objective:

Case Situation

Required

Final Answer

SECTION III

FCFF, FCFE, and Explicit-Period Cash Flows

Case 18: Estimating FCFF from Operating Profit

Learning Objective:

Case Situation

Required

Final Answer

Case 19: Estimating FCFE from Net Income

Learning Objective:

Case Situation

Required

Final Answer

Case 20: Reconciliation between FCFF and FCFE

Learning Objective:

Case Situation

Required

Final Answer

Case 21: Levered versus Unlevered Free Cash Flow

Learning Objective:

Case Situation

Required

Final Answer

Case 22: Forecasting Intermediate-Period Revenue and Margin

Learning Objective:

Case Situation

Required

Final Answer

Case 23: Growth, Reinvestment, and Free Cash Flow

Learning Objective:

Case Situation

Required

Final Answer

Case 24: Positive FCFF but Negative FCFE

Learning Objective:

Case Situation

Required

Final Answer

Case 25: Choosing FCFF under Changing Leverage

Learning Objective: 

Case Situation

Required

Final Answer

Case 26: FCFE Valuation of a Bank or NBFC

Learning Objective:

Case Situation

Required

Final Answer

Case 27: FCFF Valuation and the Equity-Value Bridge

Learning Objective:

Case Situation

Required

Final Answer

SECTION IV

Terminal Cash Flow, Terminal Value, and Long-Term Growth

Case 28: Normalising Terminal-Year Cash Flow

Learning Objective:

Case Situation

Required

Final Answer

Case 29: Perpetuity-Growth Terminal Value

Learning Objective:

Case Situation

Required

Final Answer

Case 30: Exit-Multiple Terminal Value

Learning Objective:

Case Situation

Required

Final Answer

Case 31: Long-Term Growth and the Indian Economy

Learning Objective:

Case Situation

Required



Final Answer

Case 32: Transition from High Growth to Stable Growth

Learning Objective:

Case Situation

Required

Final Answer

Case 33: Terminal-Value Sensitivity

Learning Objective:

Case Situation

Required

Final Answer

SECTION V

P/E Valuation and PEG Analysis

Case 34: Stable-Growth Justified P/E

Learning Objective:

Case Situation

Required

Final Answer

?

Case 35: Two-Stage P/E Valuation

Learning Objective:

Case Situation

?

Required

?

Final Answer

?

?

Case 36: Retention Ratio, ROE, and P/E

Learning Objective:

Case Situation

Required

Final Answer

Case 37: Trailing versus Forward P/E

Learning Objective: ?

Case Situation

?

?

Required

Final Answer

?

?

Case 38: Negative Earnings and the Limitation of P/E

Learning Objective:

Case Situation

Required

Final Answer



Case 39: PEG Ratio for a Growth Company

Learning Objective:

Case Situation

Required

Final Answer

Case 40: Peer P/E and Investment Recommendation

Learning Objective:

Case Situation



Required

Final Answer

SECTION VI

P/B Valuation

Case 41: Stable-Growth Justified P/B

Learning Objective:

Case Situation

Required

Final Answer

?

Case 42: Two-Stage P/B Valuation

Learning Objective:

Case Situation

Required

Final Answer

Case 43: ROE-Cost of Equity Spread

Learning Objective:

Case Situation

Required

Final Answer

Case 44: P/B Valuation of a Bank

Learning Objective:

Case Situation

Required

Final Answer

Case 45: Accounting Limitations of P/B

Learning Objective:

Case Situation

Required

Final Answer

SECTION VII

P/S and P/CF Valuation

Case 46: Stable-Growth Justified P/S

Learning Objective: ?

Case Situation

?

Required

?

?

Final Answer

?

?

Case 47: Two-Stage P/S Valuation**Learning Objective:**

?

Case Situation

?

Required

?

Final Answer

?

Case 48: Margin and P/S Interpretation Matrix**Learning Objective:**

?

Case Situation

?

?

?

?

?

Required

?

?

Final Answer

?

?

?

?

?

Case 49: P/S for a Loss-Making Digital Business**Learning Objective:**

?

Case Situation

?

Required

?

Final Answer

?

Case 50: P/CF and Earnings Quality**Learning Objective:****Case Situation**

?

Required

Final Answer

?

SECTION VIII

Enterprise Value, Non-Operating Assets, Beta, and APV

Case 51: Enterprise Value Bridge

Learning Objective:

Case Situation

Required

Final Answer

Case 52: Valuing Excess Cash and Marketable Securities

Learning Objective: ?

Case Situation

Required

Final Answer

Case 53: Cross-Holdings and Subsidiary Investments

Learning Objective:

Case Situation

Required

Final Answer

Case 54: Surplus Land and Idle Assets

Learning Objective:

Case Situation

Required

Final Answer

Case 55: Unlevering the Beta

Learning Objective:

Case Situation

?

Required

Final Answer

Case 56: Relevering the Beta for a Target Capital Structure

Learning Objective:

Case Situation

Required

Final Answer

Case 57: Adjusted Present Value under Changing Debt

Learning Objective:

Case Situation

Required

Final Answer

SECTION IX

EVA, MVA, and Value Creation

Case 58: Calculating Economic Value Added

Learning Objective:

Case Situation

Required



Final Answer

Case 59: Comparing Accounting Profit and EVA

Learning Objective:

Case Situation

Required

Final Answer

Case 60: Market Value Added

Learning Objective:

Case Situation

Required

Final Answer

Case 61: Linking EVA to Business Decisions and MVA

Learning Objective:

Case Situation

Required

Final Answer

SECTION X

LBO and Integrated Valuation Applications

Case 62: LBO Sources, Uses, and Debt Capacity

Learning Objective:

Case Situation

Required

Final Answer



Case 63: Cash Sweep and Debt Repayment

Learning Objective:

Case Situation

Required

Final Answer

Case 64: Exit Value and Sponsor IRR

Learning Objective:

Case Situation

Required

Final Answer

Case 65: Integrated Equity Valuation and Recommendation

Learning Objective:

Case Situation

?

Required

Final Answer

Summary of Principal Valuation Formulae

Valuation Gap %:

Equity Value Bridge:

Normalised PAT:

Operating Cash Flow (Indirect Method):

Free Cash Flow (CFO - Capex):

CF/Discount-Rate Matching:

Constant-Growth DDM:

Sustainable Growth ($g=b \times ROE$):

PVGO:

Justified Value via $g=b \times ROE$:

Two-Stage DDM:

Transition-Period DDM:

Three-Stage DDM:

Multistage DDM from EPS/Payout:

Naive Zero-Growth DDM:

$FCFF = EBIT(1-t) + Dep - Capex - \Delta WC$:

$FCFE = NI + Dep - Capex - \Delta WC + NetBorrowing$:

$FCFE = FCFF - Int(1-t) + NetBorrowing$:

FCFF & FCFE (levered vs unlevered):

Explicit-Period FCFF Forecast:

NOPAT & FCFF Forecast:

FCFE with Net Debt Repayment:

FCFF/WACC vs FCFE/ K_e Selection:

FCFE (Distributable, Regulatory Capital):

EV via FCFF, Terminal Value, Equity Bridge:

Normalised Terminal Cash Flow:

Perpetuity-Growth Terminal Value:

Exit-Multiple Terminal Value:

Long-Term Growth Sanity Check:

Mature-Stage Cash-Flow Rebuild:

Terminal Value Sensitivity (WACC, g):

Justified P/E:

Two-Stage Justified P/E:

$g = b \times ROE$ & Justified P/E:

Trailing/Forward P/E:

P/E Inapplicability (Negative EPS):

$PEG = P/E \div \text{Growth}$:

Peer P/E Adjustment:

Justified P/B = $(ROE - g) / (K_e - g)$:

Two-Stage Justified P/B:

Justified P/B (ROE- K_e spread):

Justified P/B & Target Price:

P/B Conceptual Limitations:

Justified P/S:

Two-Stage Justified P/S:

P/S-Margin Matrix:

P/S for Loss-Making Business:

$P/CF = \text{Price} / \text{OCF per share}$:

Enterprise Value Bridge:

Excess Cash Calculation:

Cross-Holding Value with Discount:

After-Tax Realisable Value:

Unlevered Beta:

Relevered Beta & Cost of Equity (CAPM):

Adjusted Present Value (APV):

$EVA = NOPAT - (\text{Invested Capital} \times WACC)$:

$MVA = \text{Market Value} - \text{Invested Capital}$:

Incremental EVA:

LBO Sources & Uses:

Cash Sweep / Debt Paydown:

Exit Equity Value, Money Multiple, IRR:

Weighted Target Price (Multi-Method):

Guide for Selecting an Appropriate Valuation Method

?

?

Common Equity-Valuation Errors

?

Sensitivity-Analysis Guide

Glossary of Important Valuation Terms

Intrinsic Value:

Enterprise Value:

Equity Value:

Sustainable Growth Rate:

Terminal Value:

Free Cash Flow to the Firm (FCFF):

Free Cash Flow to Equity (FCFE):

Justified Multiple:



PVGO:

Economic Value Added (EVA):

Market Value Added (MVA):

Adjusted Present Value (APV):

Topic-to-Case Index

Intrinsic value vs. market price:

Equity value vs. enterprise value:

Normalized/sustainable earnings:

Operating cash flow & earnings quality:

Free cash flow reconstruction:

Matching cash flow and discount rate:

Constant-growth (Gordon) DDM:

Sustainable growth ($g = b \times ROE$):

Present value of growth opportunities (PVGO):

Retention ratio and value creation:

Dividend payout policy:

Two-stage DDM:

Transition-period growth models:

Three-stage DDM:

Multistage dividends from EPS/payout:

DDM suitability and limitations:

Valuation sensitivity to K_e and g :

FCFF estimation:

FCFE estimation:

FCFF-FCFE reconciliation:

Levered vs. unlevered cash flow:

Growth and reinvestment effects on FCF:

Choosing FCFF vs. FCFE under changing leverage:

Terminal cash flow normalisation:

Perpetuity-growth terminal value:

Exit-multiple terminal value:

Long-term growth sustainability:

Justified P/E:

Trailing vs. forward P/E:

P/E limitations (negative earnings):

PEG ratio:

Peer/relative P/E multiples:

Justified P/B:

ROE-cost of equity spread:

P/B accounting limitations:

Justified P/S:

P/S and margin interpretation:

P/CF and earnings quality:

Enterprise value bridge:

Excess cash and non-operating assets:

Cross-holdings / holding-company discount:

Surplus land / idle assets:

Unlevering beta:

Relevering beta:

Adjusted Present Value (APV):

Economic Value Added (EVA):

Market Value Added (MVA):

LBO sources and uses:

Cash sweep and debt paydown:

Exit value and sponsor IRR:

Integrated / reconciled valuation:

Industry-Context Index

Auto components / auto-ancillary:

Logistics:

Textiles:

Consumer durables:

Packaging:

Renewable energy / solar:

FMCG:

Paints:

Pharmaceuticals:

Diagnostics:

Electric vehicle (EV) components:

Apparel:

Speciality chemicals:

Engineering:

Hospitals / healthcare:

Cement:

Consumer finance / NBFC:

Infrastructure / roads:

Organised retail:

Telecom:

Digital payments:

Steel:

Food delivery / digital platforms:

IT services:

Housing finance:

Banking / private lender:

Financial services:

Food processing:

Pharmacy retail:

Hotels / hospitality:

Household products:

Auto dealership:

Building materials:

PSU / public sector:

Holding companies:

Real estate / land:

Private equity / LBO:

Consumer electronics:

Formula Index

Valuation Gap %:

Equity Value Bridge:

Normalised PAT:

Operating Cash Flow (Indirect Method):

Free Cash Flow (CFO - Capex):

CF/Discount-Rate Matching:

Constant-Growth DDM:

Sustainable Growth ($g=b \times ROE$):

PVGO:

Justified Value via $g=b \times ROE$:

Two-Stage DDM:

Transition-Period DDM:

Three-Stage DDM:

Multistage DDM from EPS/Payout:

Naive Zero-Growth DDM:

FCFF = $EBIT(1-t) + Dep - Capex - \Delta WC$:

FCFE = $NI + Dep - Capex - \Delta WC + NetBorrowing$:

FCFE = $FCFF - Int(1-t) + NetBorrowing$:

FCFF & FCFE (levered vs unlevered):

Explicit-Period FCFF Forecast:

NOPAT & FCFF Forecast:

FCFE with Net Debt Repayment:

FCFF/WACC vs FCFE/Ke Selection:

FCFE (Distributable, Regulatory Capital):

EV via FCFF, Terminal Value, Equity Bridge:

Normalised Terminal Cash Flow:

Perpetuity-Growth Terminal Value:

Exit-Multiple Terminal Value:

Long-Term Growth Sanity Check:

Mature-Stage Cash-Flow Rebuild:

Terminal Value Sensitivity (WACC, g):

Justified P/E:

Two-Stage Justified P/E:

$g = b \times ROE$ & Justified P/E:

Trailing/Forward P/E:

P/E Inapplicability (Negative EPS):

$PEG = P/E \div \text{Growth}$:

Peer P/E Adjustment:

Justified P/B = $(ROE - g) / (K_e - g)$:

Two-Stage Justified P/B:

Justified P/B (ROE- K_e spread):

Justified P/B & Target Price:

P/B Conceptual Limitations:

Justified P/S:

Two-Stage Justified P/S:

P/S-Margin Matrix:

P/S for Loss-Making Business:

$P/CF = \text{Price} / \text{OCF per share}$:

Enterprise Value Bridge:

Excess Cash Calculation:

Cross-Holding Value with Discount:

After-Tax Realisable Value:

Unlevered Beta:

Relevered Beta & Cost of Equity (CAPM):

Adjusted Present Value (APV):

$EVA = NOPAT - (\text{Invested Capital} \times WACC)$:

$MVA = \text{Market Value} - \text{Invested Capital}$:

Incremental EVA:

LBO Sources & Uses:

Cash Sweep / Debt Paydown:

Exit Equity Value, Money Multiple, IRR:

Weighted Target Price (Multi-Method):